Practical-10

Demonstrate **Node.js** using **MongoDB** to perform **Create, Read, Update, and Delete operations.**

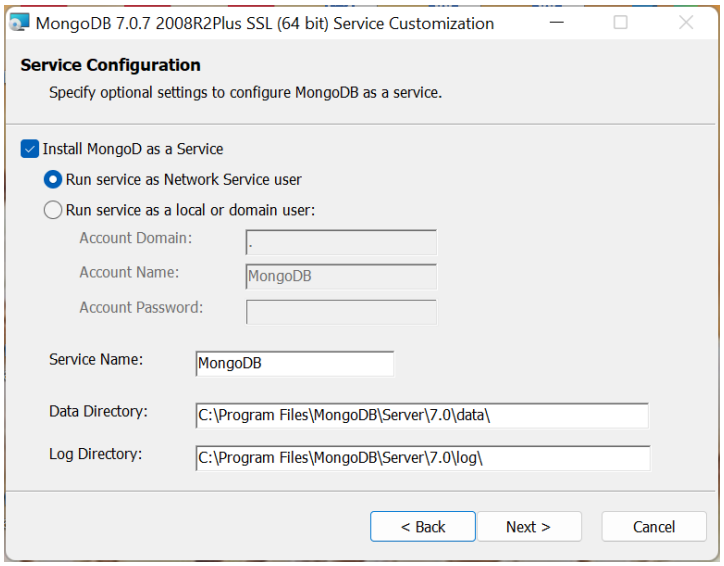
To perform CRUD operations in MongoDB using Node.js in VS Code, you'll first need to set up your environment and code editor properly. Here's a step-by-step guide**:**

Step-1. Set Up MongoDB**:** Ensure that MongoDB is installed and running on your local machine. You can download MongoDB from the official website

https://www.mongodb.com/try/download/community (Mandatory)

MongoDB Compass Download (GUI) | MongoDB (optional)

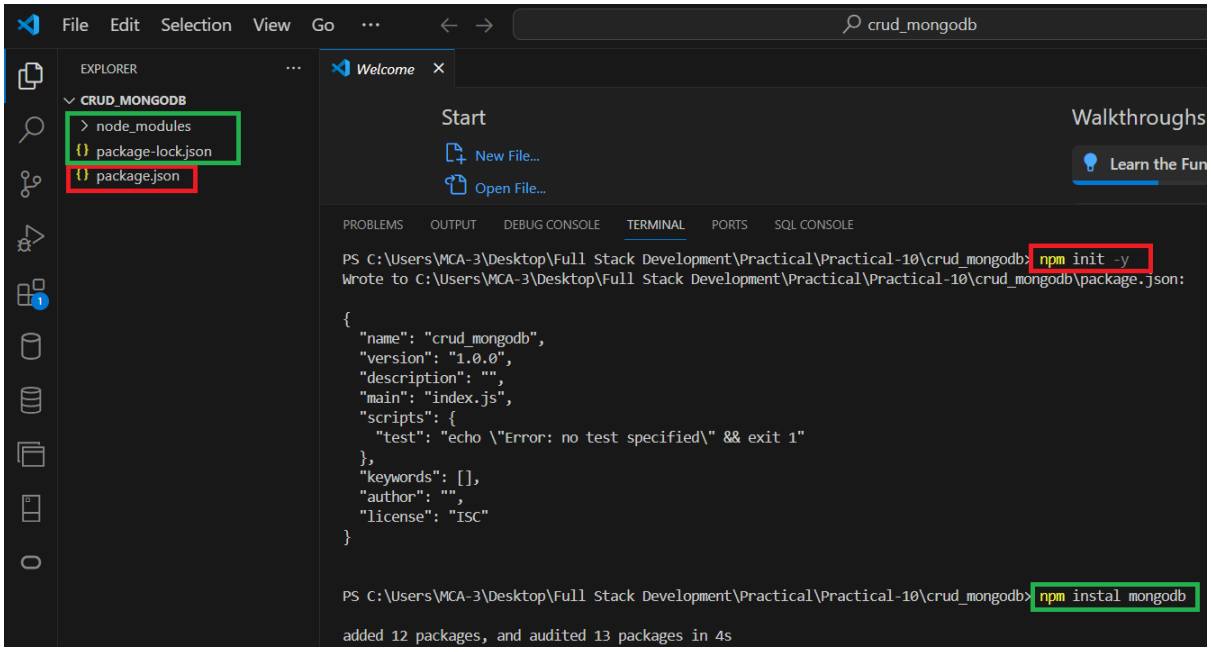
https://www.mongodb.com/try/download/shell (optional)



Step-2. Create a New Node.js Project**:** Open VS Code and create a new folder for your Node.js project. Open this folder in VS Code**.**

Step-3. Initialize Node.js Project**:** Open the integrated terminal in VS Code and navigate to your project folder. Then run the following command to initialize a new Node.js project**:**

Step-4. Install Required Packages**:** Install the mongodb package using npm. This package allows Node.js applications to interact with MongoDB.



Step-5. Create JavaScript Files**:** Create separate JavaScript files for each CRUD operation (create, read, update, delete).

File name: create.js

```
const { MongoClient } = require('mongodb');

async function main() {
  const uri = 'mongodb://localhost:27017';
  const client = new MongoClient(uri);

  try {
    await client.connect();

    const database = client.db('mydatabase');
    const collection = database.collection('mycollection');

    const result = await collection.insertOne({ name: 'John', age:
30, city: 'New York' });
    console.log(`Document inserted with id: ${result.insertedId}`);
  } finally {
    await client.close();
  }
}

main().catch(console.error);
```

File name: read.js

```
const { MongoClient } = require('mongodb');

async function main() {
  const uri = 'mongodb://localhost:27017';
  const client = new MongoClient(uri);

  try {
    await client.connect();

    const database = client.db('mydatabase');
    const collection = database.collection('mycollection');

    const documents = await collection.find({}).toArray();
    console.log('Documents:');
    console.log(documents);
  } finally {
    await client.close();
  }
}

main().catch(console.error);
```

File name: update.js

```
async function main() {
  const uri = 'mongodb://localhost:27017';
  const client = new MongoClient(uri);

  try {
    await client.connect();

    const database = client.db('mydatabase');
    const collection = database.collection('mycollection');

    const result = await collection.updateMany({ name: 'John' }, {
$set: { age: 35 } });
    console.log(`${result.modifiedCount} documents updated.`);
  } finally {
    await client.close();
  }
}

main().catch(console.error);
```

File name: delete.js

```
const { MongoClient } = require('mongodb');

async function main() {
  const uri = 'mongodb://localhost:27017';
  const client = new MongoClient(uri);

  try {
    await client.connect();

    const database = client.db('mydatabase');
    const collection = database.collection('mycollection');

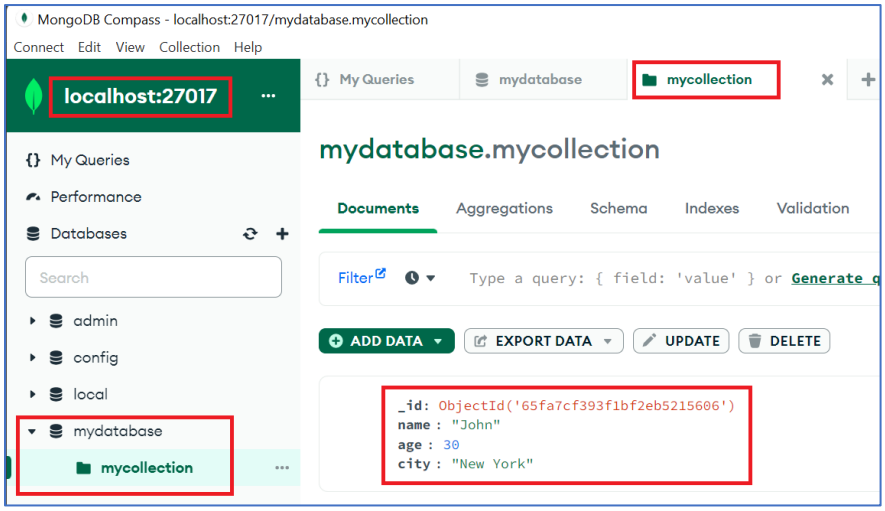
    const result = await collection.deleteMany({ name: 'John' });
    console.log(`${result.deletedCount} documents deleted.`);
  } finally {
    await client.close();
  }
}

main().catch(console.error);
```

Step-6. Run the script: Save your create.js file and return to the terminal in Visual Studio Code. Run the script by executing the following command:

```
node create.js
node read.js
node update.js
node delete.js
```

Step-7. Verify the output**:** Check the output in the terminal to see if the script executed successfully.


MongoDB Compass - localhost:27017/mydatabase.mycollection
Connect Edit View Collection Help
localhost:27017
My Queries
Performance
Databases
Search
admin
config
local
mydatabase
mycollection
My Queries
mydatabase
mycollection
mydatabase.mycollection
Documents
Aggregations
Schema
Indexes
Validation
Filter
Type a query: { field: 'value' } or Generate q
ADD DATA
EXPORT DATA
UPDATE
DELETE
_id: ObjectId('65fa7cf393f1bf2eb5215606')
name : "John"
age : 30
city : "New York"